IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

Corrected Copy

FIELDS

Tuesday, 28 May 10:00 am

Time allowed: 1:30 hours

There are THREE questions on this paper.

Question One carries 40 marks. Question Two and Question Three carry 20 marks each.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	R.R.A. Syms
	Second Marker(s) :	S. Lucyszyn

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Electromagnetic Fields 2019 – Formula Sheet

- **Differential operators (Cartesian co-ordinates)**

$\nabla = \partial/\partial x\ \underline{i} + \partial/\partial y\ \underline{j} + \partial/\partial z\ \underline{k}$

$\nabla\phi = \partial\phi/\partial x\ \underline{i} + \partial\phi/\partial y\ \underline{j} + \partial\phi/\partial z\ \underline{k}$

$\nabla \cdot \underline{\mathbf{F}} = \partial \mathbf{F}_x/\partial x + \partial \mathbf{F}_y/\partial y + \partial \mathbf{F}_z/\partial z$

$\nabla \times \underline{\mathbf{F}} = \{\partial \mathbf{F}_z/\partial y - \partial \mathbf{F}_y/\partial z\}\ \underline{i} + \{\partial \mathbf{F}_x/\partial z - \partial \mathbf{F}_z/\partial x\}\ \underline{j} + \{\partial \mathbf{F}_y/\partial x - \partial \mathbf{F}_x/\partial y\}\ \underline{k}$

$\nabla^2\phi = \partial^2\phi/\partial x^2 + \partial^2\phi/\partial y^2 + \partial^2\phi/\partial z^2$

- **Identity**

$\nabla \times \nabla \times \underline{\mathbf{F}} = \nabla(\nabla \cdot \underline{\mathbf{F}}) - \nabla^2\underline{\mathbf{F}}$

- **Integral theorems**

$\iint_A \underline{\mathbf{F}} \cdot d\underline{a} = \iiint_V \nabla \cdot \underline{\mathbf{F}}\ dv$ (Gauss' theorem)

$\int_L \underline{\mathbf{F}} \cdot d\underline{L} = \iint_A (\nabla \times \underline{\mathbf{F}}) \cdot d\underline{a}$ (Stokes' theorem)

- **Maxwell's equations – integral form**

$\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \rho\ dv$ (Gauss' Law)

$\iint_A \underline{\mathbf{B}} \cdot d\underline{a} = 0$ (Magnetic equivalent of Gauss' law)

$\int_L \underline{\mathbf{E}} \cdot d\underline{L} = -\iint_A \partial\underline{\mathbf{B}}/\partial t \cdot d\underline{a}$ (Faraday's law)

$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$ (Ampère's Law)

- **Maxwell's equations – differential form**

$\mathrm{div}(\underline{\mathbf{D}}) = \rho$

$\mathrm{div}(\underline{\mathbf{B}}) = 0$

$\mathrm{curl}(\underline{\mathbf{E}}) = -\partial\underline{\mathbf{B}}/\partial t$

$\mathrm{curl}(\underline{\mathbf{H}}) = \underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t$

- **Constitutive equations**

$\underline{\mathbf{J}} = \sigma\,\underline{\mathbf{E}}$

$\underline{\mathbf{D}} = \varepsilon\,\underline{\mathbf{E}}$

$\underline{\mathbf{B}} = \mu\,\underline{\mathbf{H}}$

- **Electromagnetic waves (pure dielectric media)**

Time dependent vector wave equation $\nabla^2\underline{\mathbf{E}} = \mu_0\varepsilon\,\partial^2\underline{\mathbf{E}}/\partial t^2$

Time independent scalar wave equation $\nabla^2\underline{E} = -\omega^2\mu_0\varepsilon_0\varepsilon_r\,\underline{E}$

For z-going, x-polarized plane waves $d^2E_x/dz^2 + \omega^2\mu_0\varepsilon_0\varepsilon_r E_x = 0$

Where $\underline{E}$ is a time-independent vector field

Plane wave solutions have the form $E(z) = E_0 \exp(-jk_0z)$

Where $k = \omega\sqrt{(\mu_0\varepsilon_0\varepsilon_r)}$ is the propagation constant

Impedance of free space $Z_0 = \sqrt{(\mu_0/\varepsilon_0)}$

Refractive index $n = \sqrt{\varepsilon_r}$

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- **Power**

Instantaneous power flow $\underline{S} = \underline{E} \times \underline{H}$

Time-averaged power flow $\underline{S} = 1/2\ \mathrm{Re}(\underline{E} \times \underline{H}^*)$

- **Transmission line formulae**

Transmission line equations for line with per unit length inductance L_p and capacitance C_p

$dV/dz = -j\omega L_p I$

$dI/dz = -j\omega C_p V$

Phase velocity and characteristic impedance of lossless line with per unit length inductance L_p and capacitance C_p

$v_{ph} = 1/\sqrt{(L_p C_p)}$

$Z_0 = \sqrt{(L_p/C_p)}$

Reflection and transmission coefficients at junction between lines of impedance Z_1 and Z_2

$R_V = (Z_2 - Z_1)/(Z_2 + Z_1)$

$T_V = 2Z_2/(Z_2 + Z_1)$

Input impedance for length d of line with properties (Z_0, k) terminated by load Z_L

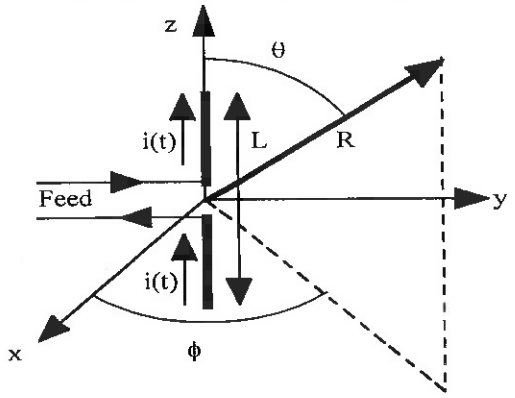
$Z_{in} = Z_0 \{Z_L + jZ_0 \tan(kd)\} / \{Z_0 + jZ_L \tan(kd)\}$

- **Antenna formulae**

Far-field pattern of half-wave dipole

$E\theta = j\ 60 I_0 \{\cos[(\pi/2)\cos(\theta)] / \sin(\theta)\} \exp(-jkR)/R$; $H_\phi = E\theta/Z_0$

Here I_0 is peak current, R is range and $k = 2\pi/\lambda$



Time averaged power flow $\underline{S} = 1/2\ \mathrm{Re}\ (\underline{E} \times \underline{H}^*) = S(R, \theta)\ \underline{r}$

Normalised radiation pattern $F(\theta, \phi) = S(R, \theta, \phi) / S_{max}$

Directivity $D = 1/ \{1/4\pi \iint_{4\pi} F(\theta, \phi) \sin(\theta)\, d\theta\, d\phi\}$

Gain $G = \eta D$ where η is antenna efficiency

Effective area $A_e = \lambda^2 D/4\pi$

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- **Transmission formulae**

Friis transmission formula $P_R = P_T \eta_T \eta_R A_T A_R/(r^2\lambda^2)$

Radar equation $P_R = P_T (\eta_T^2 A_T^2 \sigma/4\pi R^4 \lambda^2)$

- **Microwave formulae**

Skin depth $\delta = 1/\sqrt{(\pi f \mu_0 \sigma)}$

- **Optical formulae**

Snell's law $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$

Fresnel reflection coefficient $\Gamma_E = \{n_1 \cos(\theta_1) - n_2 \cos(\theta_2)\}/\{n_1 \cos(\theta_1) + n_2 \cos(\theta_2)\}$

- **Imaging formulae**

Spherical wave $E(r) = (E_0/r) \exp(-jk_0 r)$ where $r^2 = x^2 + y^2 + z^2$ in cartesian coordinates

Paraxial wave $E(R, z) = A(z) \exp(-jk_0 R^2/2z)$ where $R^2 = x^2 + y^2$ in cartesian coordinates

The lens maker's formula $1/f = (n - 1)(1/R_1 + 1/R_2)$

The imaging equation $1/u + 1/v = 1/f$

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Electromagnetic Fields 2019 – Questions

1. Using diagrams and developing formulae where appropriate, discuss briefly each of the following:

   a) Phase and group velocity

   [8]

   b) Absorption, scattering and diffraction

   [8]

   c) Interference patterns

   [8]

   d) Anti-reflection coatings

   [8]

   e) Computer aided tomography

   [8]

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2. a) In integral form, Gauss' law takes the form:

$$\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \rho \, dv$$

Using an explanatory sketch, explain the meaning of the equation. Explain briefly how Gauss' law can be used to calculate the capacitance between two electrodes.

[7]

b) Sketch the arrangement of a co-axial cable based on conductors with radii a and b, and define the Gaussian surface needed in the capacitance calculation. Assuming the conductors are held apart by plastic with a dielectric constant ε, show that the capacitance per unit length is:

$$C_p = (2\pi\varepsilon) \div \log_e(b/a)$$

[7]

c) The corresponding inductance per unit length of the cable is:

$$L_p = (\mu_0/2\pi) \log_e(b/a)$$

Calculate the propagation velocity and characteristic impedance of the cable, assuming that a = 0.5 mm, b = 2.5 mm, and $\varepsilon = \varepsilon_0\varepsilon_r$ with $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m and $\varepsilon_r = 2.25$. Comment on the value of velocity, and explain briefly why the characteristic impedance is real, if the cable is loss-less.

[6]

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3. In integral form, Ampère's law takes the form:

$$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial \underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$$

a) Explain which term was missing from Ampère's original formulation, and why it was added.

[5]

b) Using a suitable integration domain, prove that tangential components of the magnetic field must be conserved at the boundary between two different media.

[5]

c) An antenna radiates entirely isotropically in space. At a distance of 1 km, the power passing through 1 m^2 area is 5 mW. Estimate the peak electric field and the transmitter power.

[5]

d) A radar system is based on an antenna with a gain of 5000 at a frequency of 10 GHz. What is the effective area of the antenna, if it is 100% efficient? Assuming the system uses the transmitter in part (c), how much power will be received from a target with a cross-section of 20 m^2 at a distance of 10 km?

[5]

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